Computer Aided Diagnosis System for Liver Cirrhosis Based on Ultrasound Images

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ABSTRACT
This work introduces a computer-aided diagnosis (CAD) system for diagnosing liver cirrhosis in ultrasound (US) images. The proposed system uses a set of features obtained from different feature extraction methods. These features are the first order statistics (FOS), the fractal dimension (FD), the gray level co-occurrence matrix (GLCM), the Gabor filter (GF), the wavelet (WT) and the curvelet (CT) features. The measured features are presented in two different classifiers such as support vector machine (SVM) and k-nearest neighbors (K-NN). The proposed system is applied on dataset consists of 72 cirrhosis and 75 normal regions each of 128×128 pixels. The classification accuracy rates are calculated using a 10-fold cross validation. A correlation-based feature selection (CFS) is used resulting in better accuracy predictions. The results showed that SVM and K-NN classifiers achieved higher performance with the combination of the wavelet and curvelet feature vectors than other feature extraction methods.

CCS Concepts

• **Computing methodologies→Image processing.**

Keywords
Computer aided diagnosis (CAD); ultrasound images; liver diseases; feature extraction; classification.

1. INTRODUCTION
The liver is one of the biggest organs of the human body. It constitutes 2.5% of the human body weight [1]. The liver's main job is to filter the blood coming from the digestive tract, before passing it to the rest of the body. Cirrhosis is a long-term damage to the liver from several potential diseases that lead to permanent scarring. The liver becomes unable to function well. Cirrhosis develops when scar tissue replaces normal or healthy tissue in the liver. It happens after the healthy cells are damaged over a long period of time, usually many years. Early detection and quantification is important to reduce the mortality caused by liver diseases.

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Ultrasound (US) imaging procedure is one of the most widely utilized noninvasive and real-time diagnostic methods. It is used to guide doctors for diagnosing diffuse liver diseases. An ultrasound of liver uses high frequency sound waves to create a live image from inside of a patient's body. It is a painless test that is very commonly used in the medical field. A computer-aided diagnosis (CAD) system helps radiologists to diagnose the US images [2]. CAD system usually consists of segmentation of liver, extraction of features and finally identification of tissues by means of a classifier. Texture analysis of ultrasound liver images is always a challenge for researchers.

Virmani *et al.* [3] Introduced CAD system for categorizing liver into normal, cirrhotic and hepatocellular carcinoma (HCC) by using wavelet packet transform (WPT) texture descriptors, they extracted statistical features such as standard deviation, energy and mean. An accuracy of 88.8% was achieved using SVM classifier. Ahmadian *et al.* [4] Proposed a method using Gabor wavelet texture feature extraction method for categorizing different liver diseases. Features were extracted and images were classified into normal, cirrhosis and hepatitis groups using Gabor wavelet transform, dyadic wavelet transform and statistical moments. The result showed that the sensitivity is 85% in the distinction between normal and hepatitis liver images, and 86% to distinguish between normal and cirrhosis liver images. They concluded that Gabor wavelet is more appropriate than dyadic wavelet and statistical based methods.

Ribeiro *et al.* [5] identified and classified different stages of chronic liver disease. The classifiers used are SVM, K-NN, and decision tree. The best results obtained using SVM with 73.20% overall accuracy rate with a radial-basis kernel. Lee *et al.* [6] suggested feature extraction methods based on fractal geometry and spatial-frequency decomposition. Accuracies of 93.6% in distinction of cirrhosis and hepatoma and 96.7% in distinction of normal and abnormal liver were gotten by utilizing Bayes classifier. Recently, Lee [7] proposed an ensemble-based data fusion strategy to differentiate normal, hepatoma and cirrhosis. The algorithm chose satisfactory classifier with high recognition rate and diversity. An accuracy of 95.67% was accomplished. Cao *et al.* [8] extracted liver features by using 2D phase congruency to differentiate among normal, cirrhosis and fibrosis of liver. Accuracies of 96.27% for normal liver, 95% for cirrhosis and 86.6% for fibrosis were accomplished. Wu *et al.* [9] proposed two-stage feature fusion strategy to characterize ultrasonic liver tissue images in three classes: normal, hepatitis and cirrhosis. GLCM,

multi-resolution energy feature and multi-resolution fractal features were extracted. The resulting fused feature set was utilized in SVM. Accordingly, 96.25% ± 1.91 accuracy was achieved.

The rest of this paper is organized as follows. Section 2, presents the image segmentation for CAD system, the methods of feature extraction, the feature selection method and the classification methods. Section 3 describes the performance of CAD system. Section 4 summarized the experimental tests and presents the classification results. It is also discussing the results obtained using the proposed system. Finally, Conclusions and future potential research directions are proposed in Section 5.

2. MOTHODOLGY

The proposed system consists of four different steps such as region of interest (ROI) identification, feature extraction, feature selection and classification as described in the following.

2.1 Segmentation (ROI Identification)

The segmentation method decomposes an US image into small regions for further examination. Liver infections can be partitioned into two classes, diffused infections (e.g. hepatitis, cirrhosis) and focal infections (e.g. hepatoma, hemangioma). The diffuse infections are the place the variation from the norm is dispersed everywhere throughout the liver volume and the focal infections are the place the irregularity is moved in a little zone of liver tissue. Accordingly, in diffuse infections, the segmentation is any area of the image; preferably a physician determines the region that has the disease. In this work, a dataset consists of 72 ROIs for cirrhosis and 75 ROIs for normal is used. From each image, ROI with 128×128 pixels is cropped manually (see Fig. 1).

2.2 Features Extraction

For each ROI, several features have been derived from the FOS, GLCM, FD, GF, WT and CT. The first order statistics (FOS) manages the extracted data from an isolated pixel. The advantage of this method is that the features are extracted quickly. The features extracted from FOS are mean, average, energy, entropy, skewness and kurtosis [10, 11]. The gray level co-occurrence matrix extracts features based on two-pixels intensity values that different in rotational angles (0°, 45°, 90°, 135°) and distance to the neighbor pixel [12].

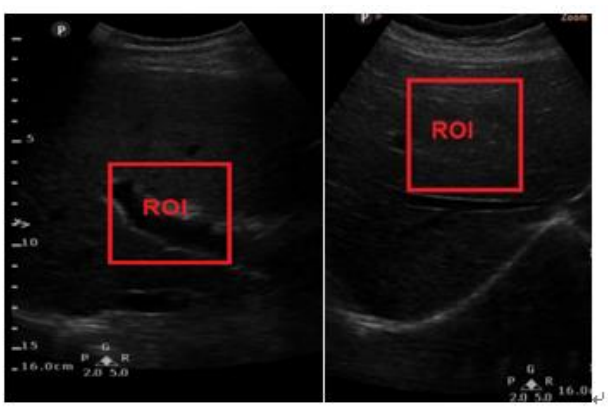

Fig. 1. Liver representation of cirrhosis (left) and normal condition (right).

The fractal dimension used to figure geometric shapes that exhibit self-similarity. Fractal dimension are calculated by box dimension strategy. The features extracted from FD are mean, standard deviation and lacunarity [13]. Gabor filter breaks down a solitary image utilizing a linear combination of various frequencies and angles. In 2D Gabor filter, a number of Gabor filter banks are applied on images that filtered by varying the wavelengths and the orientation angles, then the result is evaluated and compared for each Gabor filter bank [14].

The wavelet transform is a multi-scale analysis of an image. It has only three directional horizontal, vertical, and diagonal. The 2D wavelet transform is the mix of two 1D wavelet transform. In the first it does 1D wavelet transform along rows, and afterward do 1D wavelet transform along columns. The wavelet image decomposition provides a representation that is simple to interpret [15]. Each sub-image contains data of a specific scale and orientation, which is helpfully isolated. Spatial data is held inside the sub-images.

The curvelet transform is a multi-scale and multi-directional analysis that evolved from the wavelet transform which comprises of a few sub-bands at various scales comprising of various orientations in the frequency domain. From all these sub-bands the statistical features are computed [16-18]. Table 1 summarizes the number of features extracted from different feature extraction method.

2.3 Features Selection

The features selection (FS) method can assess individual features and rank them based on their correlation with the classes. Features selection can be utilized for reducing the computation time, the processing complexity and improving the performance of the CAD system. This work applied the correlation-based filter (CFS) approach to select the most significant features.

CFS is a straightforward filter algorithm that assorts feature subsets as indicated by a correlation based heuristic evaluation function. The inclination of the evaluation function is toward subsets that contain highlights that are profoundly related with the class, yet uncorrelated with each other [19].

CFS's feature subset evaluation function

$$M_s = \frac{k\overline{r_{ef}}}{\sqrt{k + (k-1)\overline{r_{ff}}}} \quad (1)$$

where M_s is the heuristic "merit" of a feature subset S containing k features, $\overline{r_{ef}}$ is the mean feature-class correlation ($f \in S$), and $\overline{r_{ff}}$ is the average feature-feature inter-correlation.

2.4 Classification

Image classification breaks down the numerical properties of different image features and organizes data into classes. The methods that used to classify the liver tissue are SVM and K-NN. SVM uses input features to determine a maximum margin hyper-plane to separate the training data from two classes. The special kernel function can be used in the case of the features are not linearly separable to transform the data to a higher dimensional feature space [20, 21]. On the other hand, K-NN classifier uses the k nearest neighbor's distance measure to make the decision of class attribution. The classifier is based on the majority vote of its k-nearest neighbors in a test sample. In the present work, the Euclidian distance metric is used [22].

Table 1. The number of features extracted using different features extraction methods

Features extraction methods	Number of features
FOS	6
GLCM	14
GF	7
FD	3
WT	108
CT	486

Table 2. Summary of the performance of SVM classifier method with different set of features

Feature set	TP	FP	ROC
FOS	0.871	0.129	0.873
GLCM	0.89	0.105	0.893
GF	0.782	0.221	0.781
FD	0.898	0.100	0.899
WT	0.966	0.033	0.966
CT	0.946	0.053	0.946
FOS+GLCM	0.891	0.105	0.893
FOS+GF	0.905	0.095	0.905
FOS+FD	0.966	0.033	0.966
FOS+WT	0.966	0.033	0.966
FOS+CT	0.952	0.046	0.953
GLCM+GF	0.898	0.101	0.898
GLCM+FD	0.918	0.081	0.919
GLCM+WT	0.966	0.033	0.966
GLCM+CT	0.946	0.053	0.946
GF+FD	0.939	0.060	0.939
GF+WT	0.966	0.033	0.966
GF+CT	0.993	0.007	0.993
FD+WT	0.966	0.033	0.966
FD+CT	0.952	0.046	0.935
WT+CT	0.993	0.007	0.993

3. PERFORMANC EVALUATION OF CAD SYSTEM

To determine the performance of the proposed system, some quality measures are needed to be calculated. These factors are listed as follow, accuracy, Receiver Operating Characteristic (ROC) curves, F-Measure, precision, and Area Under the Curve (AUC) [23]. The classification results are tabulated in a confusion matrix [24]. The confusion matrix would have four results: True positives (TP) are positive cases accurately classified as positive. True negatives (TN) are negative cases effectively distinguished as negatives. False positives (FP) are negative cases accurately classified as positive. False negatives (FN) are positives cases classified by the system as negative ones [25].

where: FP rate = FP / (FP + TP)

FN rate = FN/ (FN + TN)

TP rate = TP / (TP + FP)

TN rate= TN / (TN + FP)

4. RESULTS AND DISCUSSION

In this work, 147 ROIs are obtained from 72 cirrhosis images and 75 normal tissues. All cases are acquired and provided by our radiologist consultant. A region of interest (ROI) that demonstrate the focus are of liver cirrhosis is marked from each image. The ROI size is 128×128 pixels. A total of 624 features have been extracted from each ROI of liver ultrasound images, namely 6 using FOF, 14 using GLCM, 7 using GF, 3 using Fractals approach, 108 using WT, and 486 using CT. The correlation-based filter approach method has been used for obtaining the most relevant features that improved the CAD system performance. The obtained features are forwarded to the SVM and the K-NN classifiers for evaluation purposes. TP, FP and ROC are measured for each feature extraction method and the combination between some features is also tested.

The obtained results are illustrated in Tables 2 and 3. The best overall accuracy is achieved using SVM and K-NN classifiers with the combination of WT and CT feature vectors. We think that the main reason for this is that the combination of WT and CT feature vectors are extracted in more details and more texture features. We can achieve a 99.31% classification accuracy rate. A comparison demonstrates the performance of each classifiers is shown in figure 2. It is difficult to identify a classifier with higher performance from these results as the ROC values is very close in almost every case.

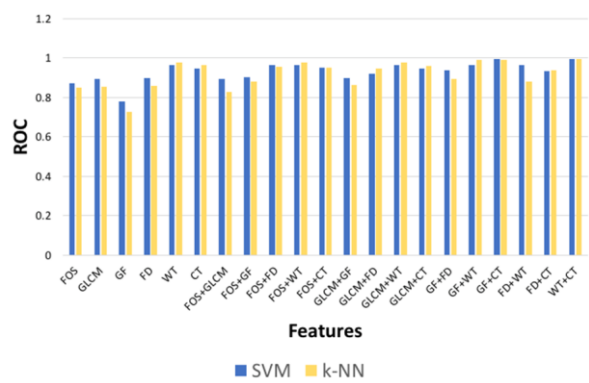

Fig. 2. Comparison of the performance of SVM and k-NN classifiers for all different features.

5. CONCLUSION

This paper proposed computer aided system for liver cirrhosis detection. Several feature extraction methods are highlighted and evaluate in order to identify the best set of features. SVM and KNN classifiers are used to accomplish the classification task. The result showed that SVM and K-NN classifiers achieved the highest classification accuracy rate with the combination of WT and CT feature vectors. It reached to 99.31%.

Since, the results were improved by applying combination of different set of features. The new directions for future work such as using some swarm intelligent methods to identifying the most important features that could improve the CAD system performance.

Table 3: Summary of the performance of KNN classifier method with different set of features

Feature set	TP	FP	ROC
FOS	0.844	0.156	0.848
GLCM	0.864	0.136	0.856
GF	0.735	0.266	0.725
FD	0.857	0.143	0.857
WT	0.980	0.021	0.979
CT	0.966	0.033	0.963
FOS+GLCM	0.837	0.163	0.829
FOS+GF	0.891	0.108	0.882
FOS+FD	0.946	0.054	0.955
FOS+WT	0.980	0.021	0.979
FOS+CT	0.959	0.039	0.952
GLCM+GF	0.878	0.123	0.864
GLCM+FD	0.946	0.054	0.948
GLCM+WT	0.980	0.021	0.979
GLCM+CT	0.959	0.039	0.958
GF+FD	0.891	0.109	0.895
GF+WT	0.993	0.007	0.989
GF+CT	0.993	0.007	0.989
FD+WT	0.891	0.108	0.882
FD+CT	0.939	0.059	0.937
WT+CT	0.993	0.007	0.993

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